



Surya as AI infrastructure

SOUTHWEST RESEARCH INSTITUTE®

Andrés Muñoz-Jaramillo and the Surya FM team

andres.munoz@swri.org

Commissioned and funded by NASA's Office of the Chief Science Data Officer (OCSDO)

Vision and leadership: Lika Guhathakurta

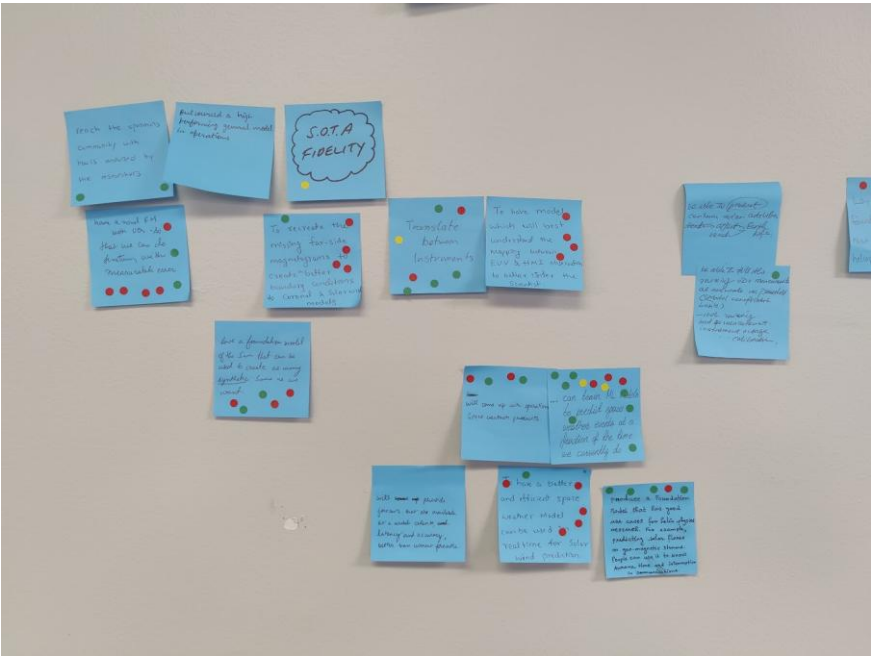


SOLAR SYSTEM SCIENCE & EXPLORATION

Infrastructure

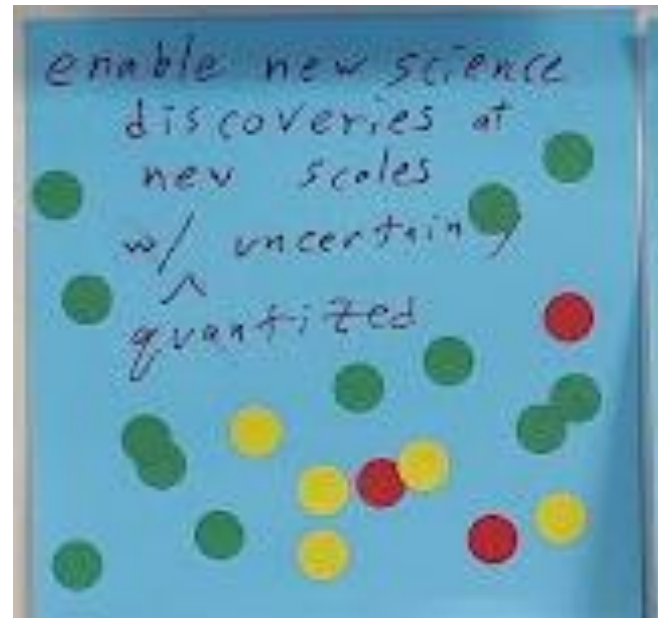
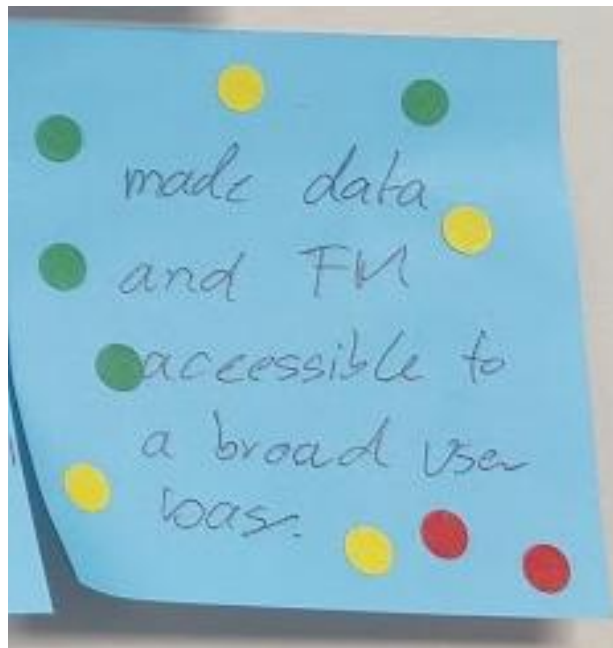
The basic physical and organizational structures and facilities (e.g. buildings, roads, power supplies) needed for the operation of a society or enterprise. It includes the foundational structures and systems, both physical (hard infrastructure) and sometimes institutional (soft infrastructure), that allow society to operate and grow

Surya Team's Goals

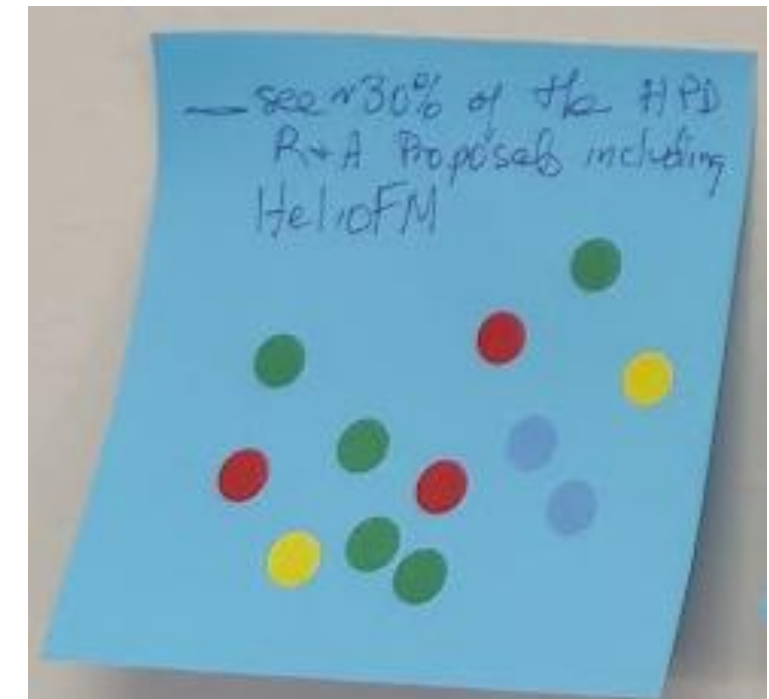
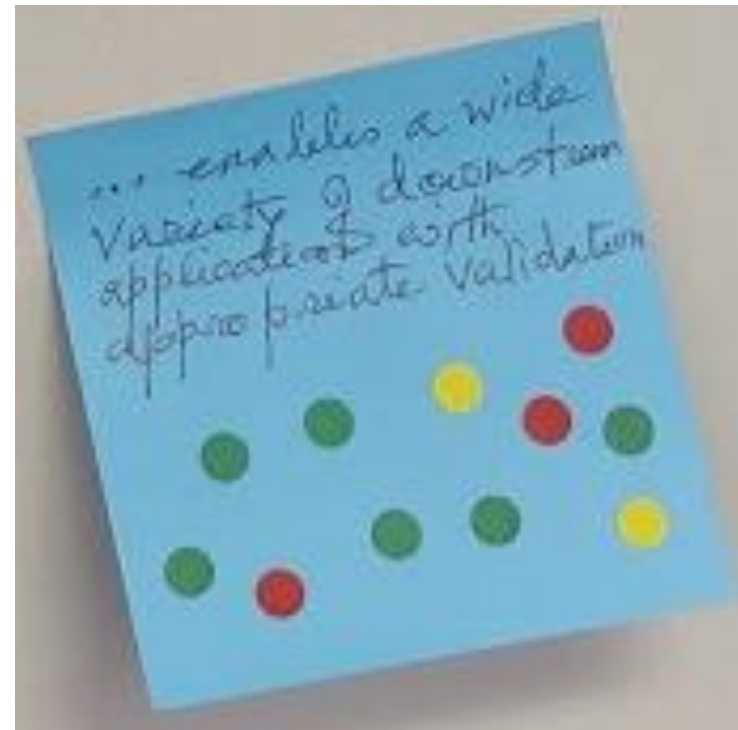


Foundational means more to us than a pretrained model

Technical



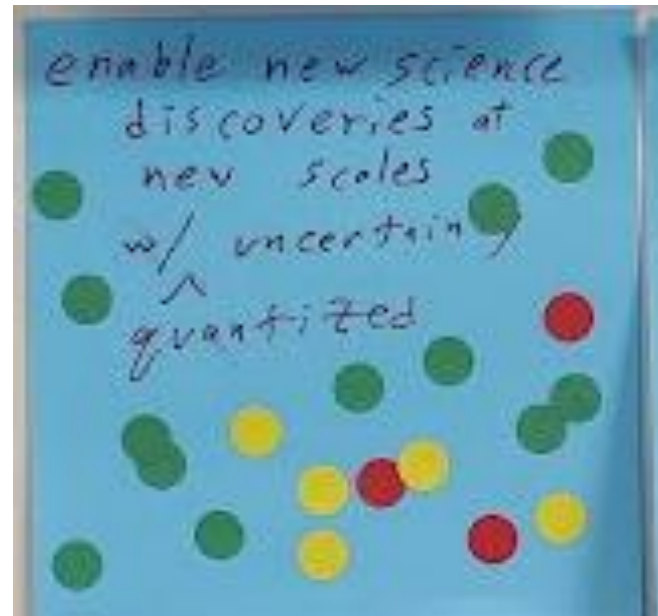
Social



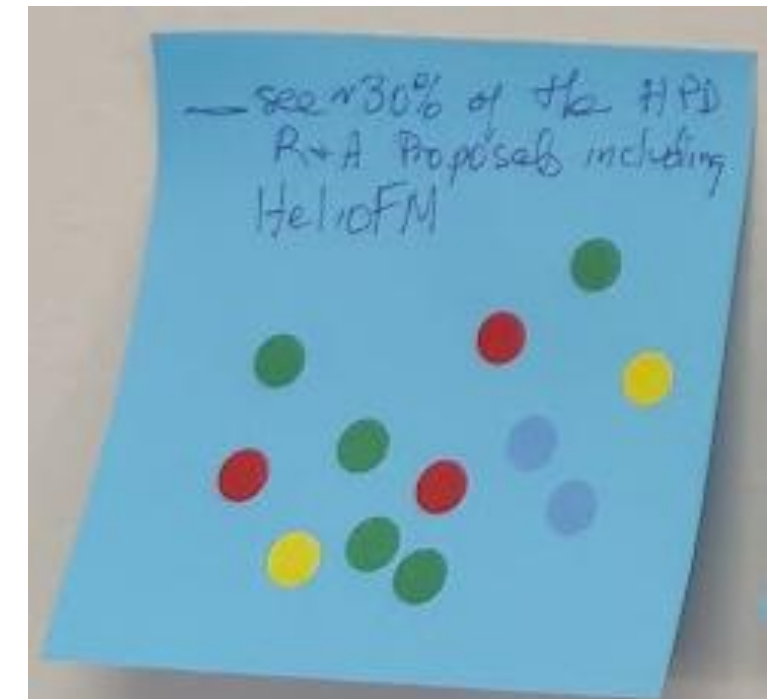
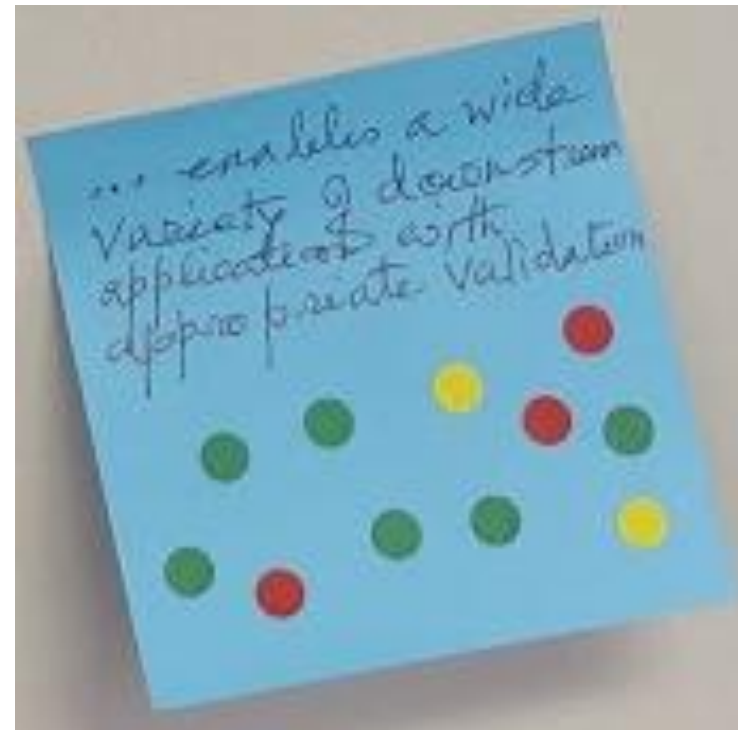
Foundational means more to us than a pretrained model

Technical

- Worked fully in a cloud environment.
- Demonstrated that Surya can be fine-tuned without creating data replicas



Social

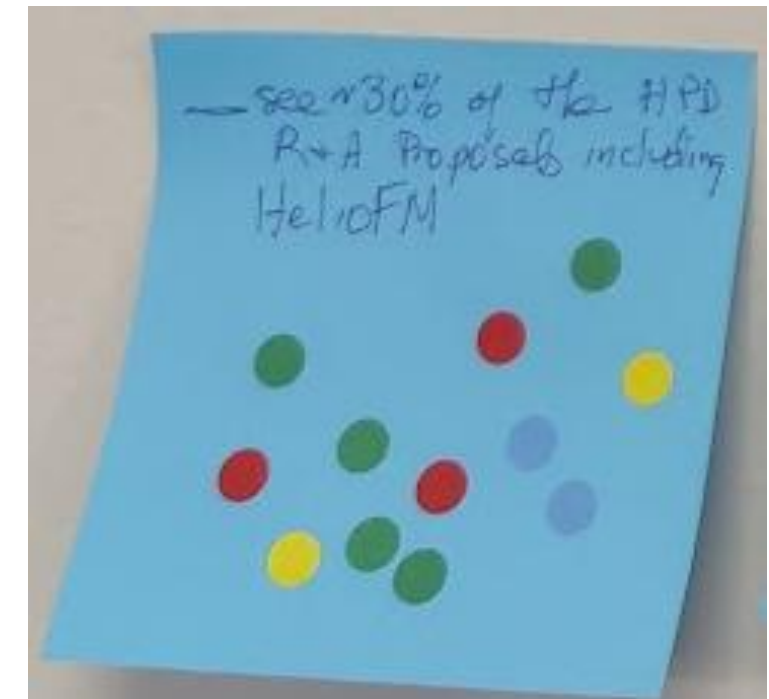
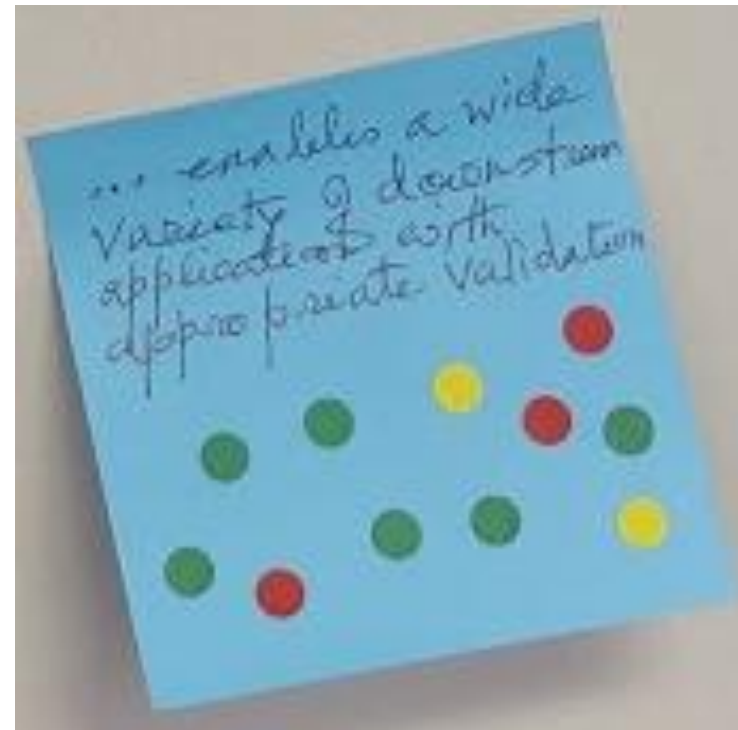


Foundational means more to us than a pretrained model

Technical

- Worked fully in a cloud environment.
- Demonstrated that Surya can be fine-tuned without creating data replicas
- Started applications that are outside of the cadence, resolution, modality, and physical domain for which Surya was trained.

Social



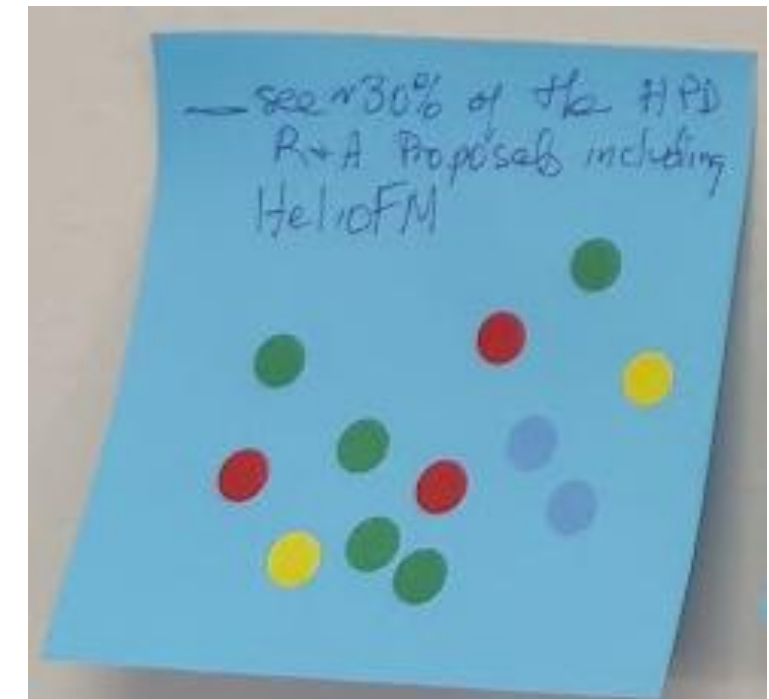
Foundational means more to us than a pretrained model

Technical

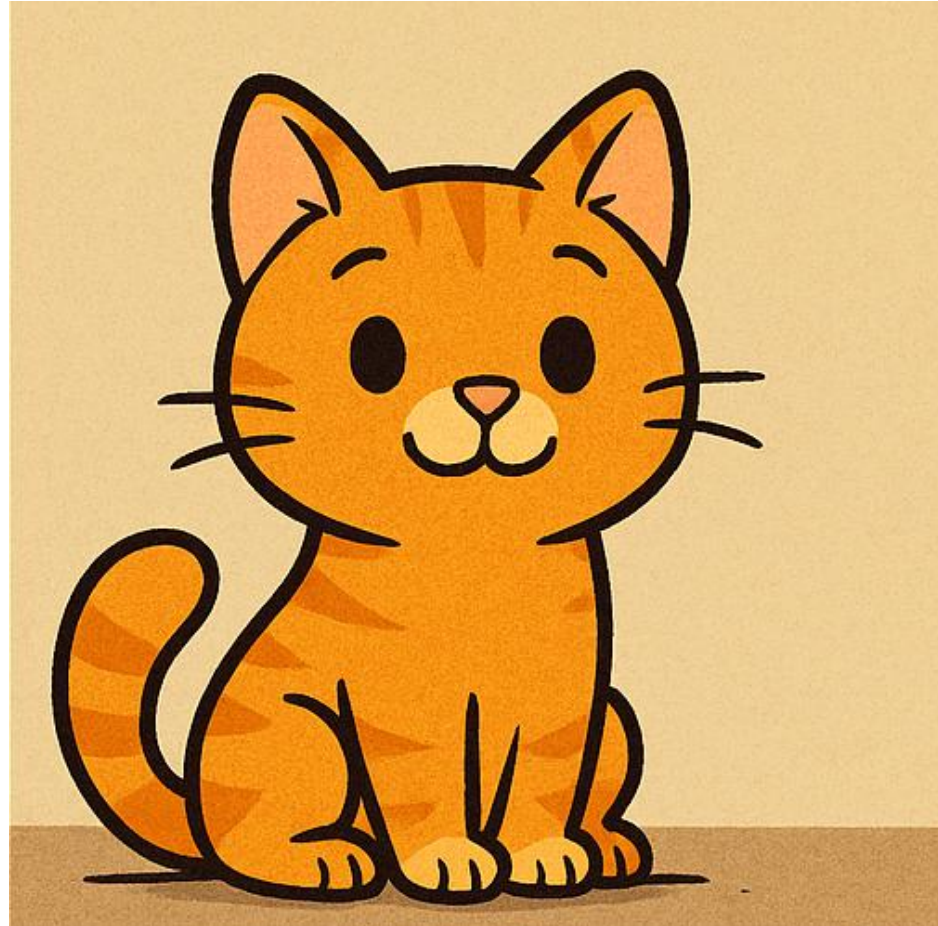
- Worked fully in a cloud environment.
- Demonstrated that Surya can be fine-tuned without creating data replicas.
- Started applications that are outside of the cadence, resolution, modality, and physical domain for which Surya was trained.

Social

- 14 teams were able to implement dataloaders, baselines and finetuning pipelines, starting only with a common template.



From zero to hero in 3 days!



Thank you so much for how graciously you took in all the jank!

Let's celebrate our progress!

